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Decoherence Noise on the Superconducting Qubits Training Program

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Introduction

Quantum computing is a growing field with promising applications in a variety of fields such as healthcare, energy consumption, and cryptography. Quantum computing leverages the principles of quantum mechanics - superposition and entanglement. These quantum phenomena allow for quantum interference and quantum parallelism. Yet, in the Noisy Intermediate Scale Quantum (NISQ) Era - quantum systems face the major challenge of decoherence due to noise. This era is characterized by low amounts of qubits and high gate error. Decoherence leads to the loss of the quantum information stored in the qubit. Noise occurs with any quantum system that is exposed to the environment. It should also be noted that quantum information can be stored in the cavity - Fermilab specializes in coupling transmons to ultrahigh-Q SRF cavities.

The Superconducting Qubits Training Program (SQTP) [1] provides a visualization for beginners in quantum computing. The open quantum system simulated is a superconducting qubit (two-level atom) coupled to a microwave cavity whose excitations are photons. The Rotating Wave Approximation of the Jaynes-Cumming Hamiltonian is used. SQTP utilizes open-source Python-based libraries scQubits, NumPy, and QuTiP alongside the Master Lindblad equation. In this project, we study the different decay behaviors of qubits and cavities with collapse operators.

Qubits

The basic unit of quantum computing is a qubit. A qubit is a “quantum bit” and refers to the basic unit of classical computing. The main principle of a qubit is that there must be two distinct states. These states can be analogous to the 0 and 1 states of a bit. A qubit can be realized physically with a wide variety of experimental quantum systems such as trapped-ions, quantum dots, and electron/nuclear spins [2]. In SQTP, the excited state and ground state are the two distinct states. The excited state is defined as when the qubit is in the $|1\rangle$ state and the cavity is in its first Fock state.

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Equation 1: The excited state of the quantum system. Tensor product between $|1\rangle$ state of qubit and first Fock state of cavity when N = 2. In QuTiP, N represents the size of the photon Hilbert space. N does not represent the Fock state.

A key distinction between a bit and qubit is that a qubit can experience superposition and entanglement. Superposition refers to the phenomenon that a qubit can be in multiple states at the same time prior to measurement. The behavior of the quantum state is described by probability amplitudes. These probability amplitudes can be negative, positive, or complex numbers. When squared, probability amplitudes become probability densities. Superposition allows for a greater amount of information to be stored and processed.

Entanglement describes how the quantum state of multiple qubits can be intertwined - allowing quantum computers to manipulate multiple qubits at once. These are the two properties which show the promise of quantum computing for simulating "highly-entangled many-particle systems." [3]. These problems are categorized as quantumly easy and classically hard. A potential application of quantum computers is in healthcare creating new pharmaceuticals for cancer and diabetes treatment [3].

Decoherence is a major challenge impacting quantum computing. Small amounts of noise can lead to decoherence. Noise occurs when the quantum system interacts with the surrounding environment. Decoherence limits the amount of useful information that can be stored and processed in the qubit. Thus, national laboratories and companies are striving for longer coherence times to extend the lifetime of a qubit – allowing for the preservation of quantum information. At Fermilab, superconducting qubits are coupled to ultrahigh-Q SRF cavities - storing quantum information in the cavity. This unique method aims to extend the coherence time of the cavity. SRF cavities are used in particle accelerators and contain superconducting material such as niobium rather than copper or aluminum found in RF cavities.

In 2018, Dr. Preskill of Caltech coined the term NISQ (Noisy Intermediate Scale Quantum) to describe the current era of quantum computing. This era is characterized with intermediate qubit scales ranging from 50 to a few hundred qubits. The NISQ era is also characterized by high error rates due to noisy gates unprotected by quantum error correction (QEC). QEC suggests that a highly entangled quantum state is necessary to preserve quantum information [3].

Qubits

This program is designed to initialize a superconducting qubit with the parameters.py section. The main.py, sweeps.py sections, [1] and sweepsnoise.py all read the parameters section. Superconducting qubits are realized physically with superconducting circuits composed of capacitors, inductors, and the Josephson Junction. A superconducting qubit is cooled to milli-Kelvin temperatures by dilution refrigerators, experiences virtually no resistance, and is macroscopic unlike non-superconducting qubits [2]. Superconducting qubits are often referred to as artificial atoms.

Open and Closed Quantum Systems

A closed quantum system is an ideal case where energy is conserved throughout time. The evolution of a closed quantum system is given by the Schrödinger equation.

$$i\hbar \frac{d|\Psi\rangle}{dt} = H|\Psi\rangle$$

Equation 2: Schrödinger equation. Closed system in terms of a pure state.

This type of system is unitary. A unitary system contains unitary operators. Unitary operators are reversible. A closed quantum system contains pure states. These pure states are often represented by kets $|\psi\rangle$ which are vectors in Dirac notation. In a closed quantum system, quantum states can then be controlled with unitary operations. The Pauli operators, Hadamard (H) gate, CNOT gate, and all quantum gates are examples of unitary operations. The application of a quantum gate twice will revert the system back to its original state.

An open quantum system considers the interaction between the qubit(s) and the surrounding environment. In this situation, the system is introduced to noise and decoherence occurs. Thus, the evolution of an open quantum system cannot be described only with unitary operations. An open quantum system contains mixed states. These mixed states are represented by density matrices. Furthermore, unlike a closed quantum system, an open quantum system cannot be represented by a ket. There must be unitary and dissipative components. Quantum master equations contain both components [4]. Solving the Lindblad master equation is one method of describing the evolution of an open quantum system.

$$\dot{p}_{tot}(t) = -\frac{i}{\hbar}[H_{tot}, \rho_{tot}(t)]$$

Equation 3: Dynamics of a closed system and the environment of a mixed state.

$$\frac{\partial \rho(t)}{\partial t} = \mathcal{L}\rho(t)$$

Equation 4: Dynamics of a closed system and the environment of a mixed state.

$$\mathcal{L}\rho(t) = -\frac{i}{\hbar}[H(t), p(t)] + \sum_n \frac{1}{2}[2C_n\rho(t)C_n^\dagger - \rho(t)C_n^\dagger C_n - C_n^\dagger C_n\rho(t)$$

Equation 5: Superoperator of the Lindblad master equation according to QuTiP documentation [7].

The unitary portion of the Lindblad master equation is the effective Hamiltonian. In this program, the Rotating Wave Approximation of the Jaynes Cumming Hamiltonian is used as the effective Hamiltonian. The dissipative portion of the Lindblad master equation is described by the superoperator L [4]. The superoperator can control density matrices transforming a mixed state to another mixed state. This program solves the Lindblad master equation with differential equations using the qutip.mesolve feature rather than with Monte Carlo methods [5].

$$H_{RWA} = \hbar\omega_c a^\dagger a + \frac{1}{2}\hbar\omega_a\sigma_z + \hbar g(a^\dagger\sigma_- + a\sigma_+)$$

Equation 6: Rotating wave approximation of the Jaynes-Cumming Hamiltonian

$\hbar = 1 \; in \; QuTiP \; program$

Jaynes-Cumming Hamiltonian

A Hamiltonian describes the energy structure of a system. The Hamiltonian is a key component of a quantum system. The Hamiltonian of a closed and open quantum system must be known to determine how the system will evolve throughout time. In a closed system, the Hamiltonian and initial state are required to solve the Schrodinger equation. In an open system, the Hamiltonian is necessary to fulfill the unitary requirement of the Lindblad master equation.

The first term is the cavity Hamiltonian, and the second term is the qubit Hamiltonian. The final term describes the interaction Hamiltonian between the cavity and qubit. A way to differentiate these terms is by the collapse operators. Collapse operators a and a† affect the cavity while collapse operators σ- and σ+ as well as the Pauli matrices affect the qubit. Thus, an interaction between the cavity and atom should contain both cavity and qubit collapse operators.

The Jaynes-Cumming Hamiltonian is the simplest model that describes light-matter interaction. This type of Hamiltonian describes the behavior of a two-level system and a harmonic oscillator [6]. In this project, the two-level system is the two-level artificial atom created by the superconducting circuit. The harmonic oscillator is a microwave cavity. This coupling is also known as cavity quantum electrodynamics (cQED).

Decoherence Noise

Decoherence noise is caused by longitudinal noise and transverse noise [2]. T1 describes the longitudinal relaxation time [2]. The longitudinal relaxation time refers to how long it takes the quantum state to decay from the excited to the ground state. T2 describes the transverse relaxation time [2]. The goal of this project is to add decoherence noise and observe longitudinal decay. “Energy decay” and “energy relaxation” are other terms used to describe longitudinal decay. Longitudinal decay is a resonant phenomenon and an irreversible process [2]. A resonant phenomenon requires noise of a certain frequency - broadband noise cannot impact the quantum system. Longitudinal decay is also described as an irreversible process because the quantum system exchanges energy with its environment and thereby loses quantum information [2].

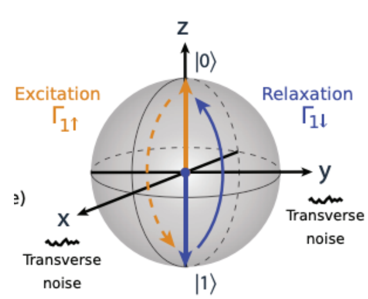

Figure 1: An illustration of longitudinal relaxation on the Bloch Sphere [2].

Decoherence noise is simulated with the addition of collapse operators. In QuTiP, collapse operators are referred to as c_ops. Common collapse operators are σ-, σ+, a, and a†. Collapse operators σ- and σ+ affect the qubit while collapse operators a and a† impact the cavity. To simulate decay in SQTP, destructive collapse operators σ- and a are used. Alongside collapse operators, the user can also drive either the qubit or cavity. In the program, the user can do so with the Hdrive class operation. The cavity is driven with a + a†a while the qubit is driven with σ- + σ+. A qubit or cavity can be driven towards the ground or excited state. In this program, the Hdrive will act in the same direction as the collapse operators and further decay the quantum system.

Resonant and Dispersive Regime

There are two main regimes where light-matter interaction can take place: the resonant and dispersive regime. The resonant limit is characterized by a strong coupling energy and small detuning of the atom and cavity [6]. Detuning describes the difference between the atom and cavity frequencies. The resonant limit arises when the difference between the atom and cavity frequency is much smaller than the coupling energy.

$$\omega_a - \omega_c << g$$

Equation 7: Resonant limit where ω_a is the atom frequency (GHz) , ω_c is the cavity frequency (GHz), and g is the coupling energy (GHz).

The dispersive limit is characterized by a weak coupling energy and large detuning of the atom and cavity [6]. Atom-cavity detuning is represented by Δ. Thus, the dispersive limit arises when the difference in atom and cavity frequencies is much greater than the coupling energy.

$$\omega_a - \omega_c \;>>\; g$$

Equation 8: Dispersive limit where ω_a is the atom frequency (GHz) , ω_c is the cavity frequency (GHz), and g is the coupling energy (GHz).

Rabi oscillations can occur in either regime. In this project, rabi oscillations are described by the Jaynes-Cumming model which contains a two-level atom and an empty cavity. Therefore, only vacuum rabi oscillations are described. Vacuum rabi oscillations are the periodic exchange between photon and atom excitation. In the resonant regime, vacuum rabi oscillations occur spontaneously [6]. In the dispersive regime rabi oscillations must be stimulated by a pulse to arise [6].

Results

The findings of this project were all taken at the dispersive limit. The rabi oscillations observed in the rabi sweeps are stimulated by square RF pulses of varying amplitude and length. In the parameter section, the atom and cavity were set to 0.001 GHz and 5.0 GHz, respectively. The coupling energy was set to 0.001 GHz - a weak coupling energy. These parameters resulted in ω_a = 0.006280 GHz and ω_c = 31.41 GHz. These frequencies were calculated by multiplying 2π by the atom and cavity parameter value. The qubit frequency is much less than the cavity frequency because the qubit is being driven by the square pulse in Figures 2-4. The lower frequency provides a lower threshold for the qubit to be driven and rabi oscillations to occur.

The collection of rabi sweeps in Figure 2 and Figure 3 showcase decay behaviors with varying kappa and gamma. Figure 2 and Figure 3 share a common trend. As a greater value of gamma or kappa is applied - the quantum system decays to the ground state at a faster rate. This observation reinforces gamma and kappa as the dissipation rates of collapse operators added onto the qubit and cavity, respectively.

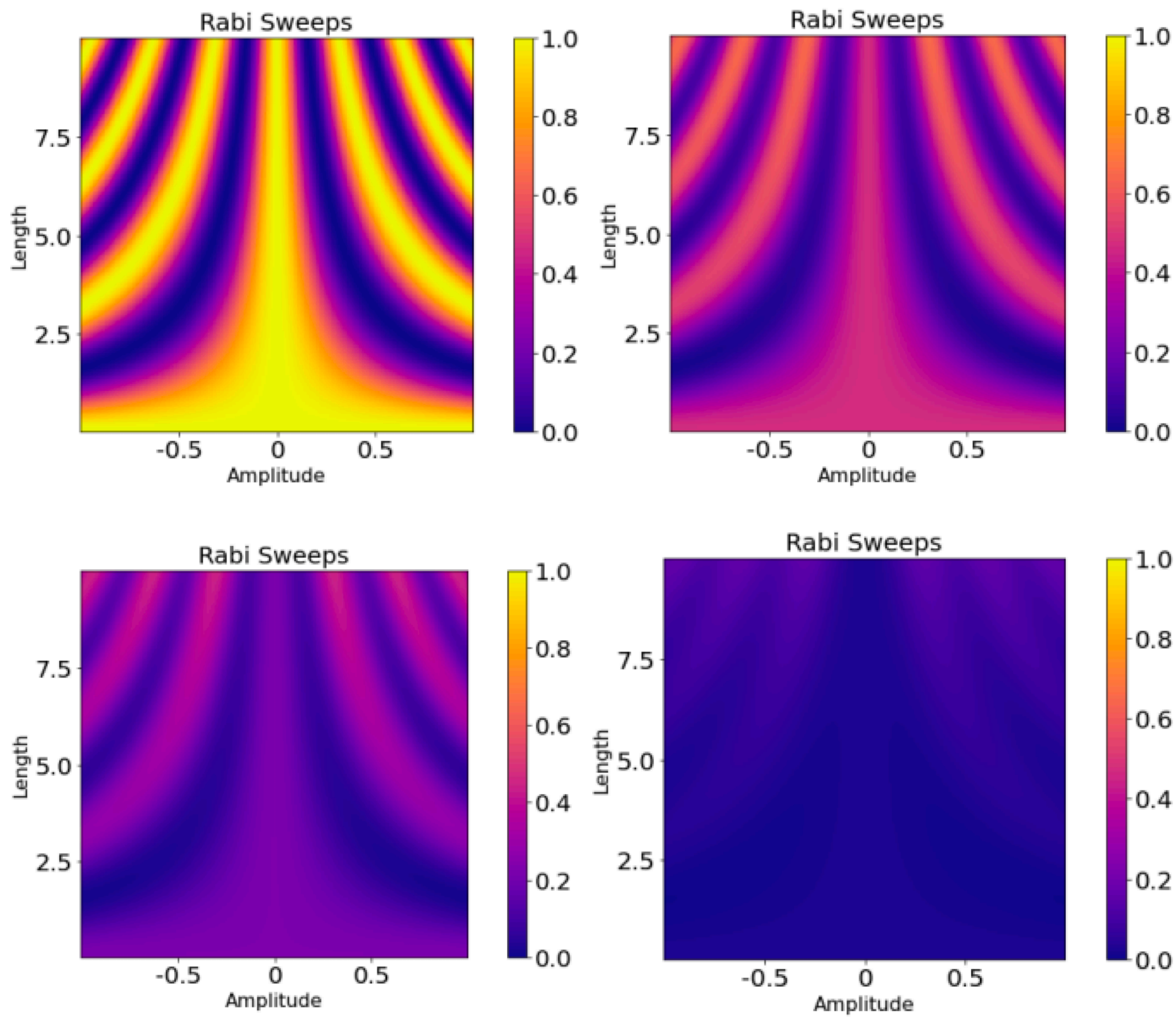

Figure 2: Rabi sweep of amplitude and length with varying gamma. The qubit, cavity, and interaction Jaynes-Cumming Hamiltonian are evolved by the Lindblad master equation. The qubit is driven by a square pulse and a σ- collapse operator is added. Gamma and kappa are qubit and cavity dissipation rates.

(a) Gamma = 0.00

(b) Gamma = 0.05

(c) Gamma = 0.10

(d) Gamma = 0.25

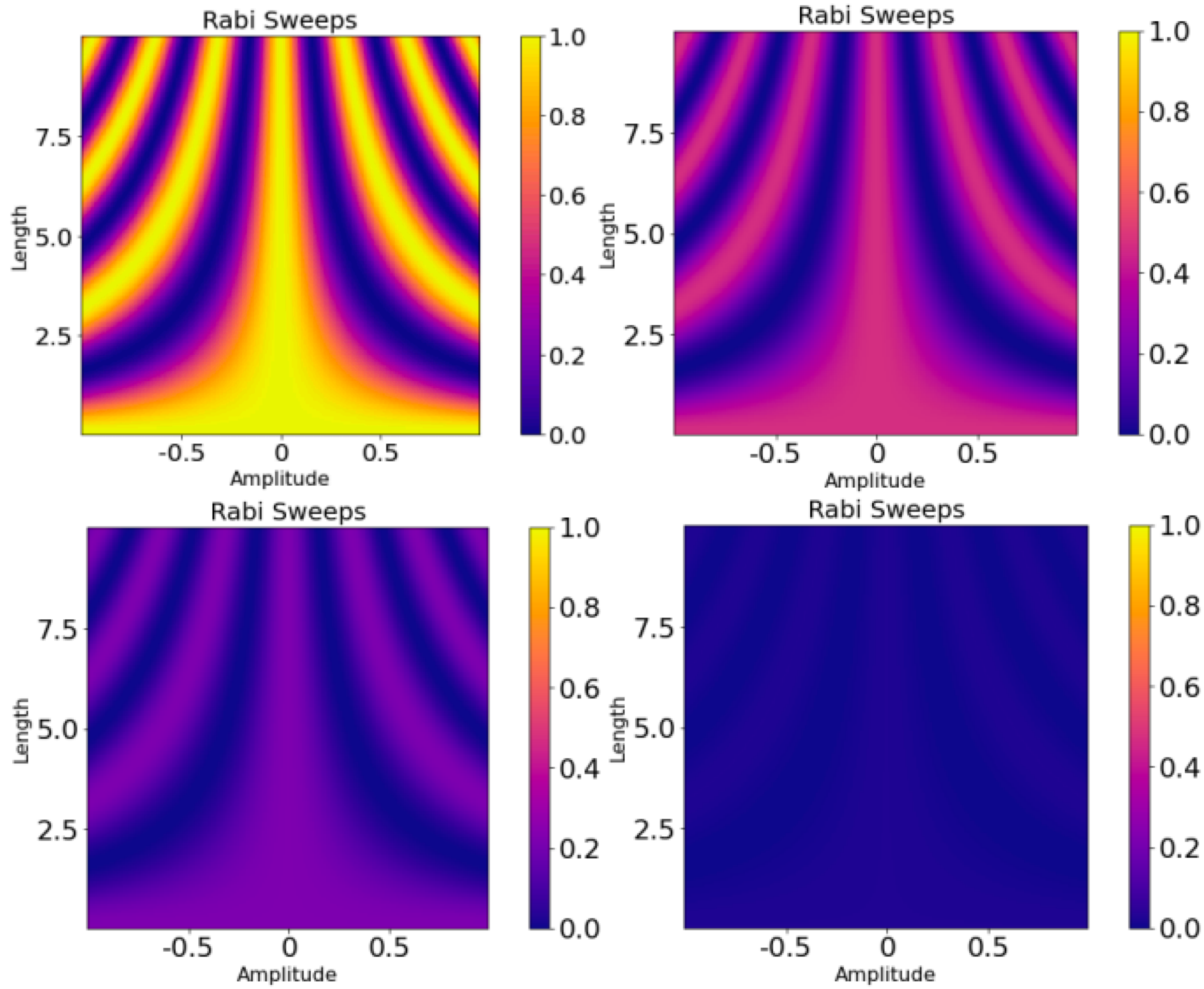

Figure 3: Rabi sweep of amplitude and length with varying kappa. The qubit, cavity, and interaction Jaynes-Cumming Hamiltonian are evolved by the Lindblad master equation. The qubit is driven by a square pulse and an annihilation collapse operator (a) is added. As the value of kappa increases, the quantum system experiences longitudinal relaxation and greater decoherence - returning to the ground state.

(a) Kappa = 0.00

(b) Kappa = 0.05

(c) Kappa = 0.10

(d) Kappa = 0.25

A difference between Figure 2 and Figure 3 is highlighted by Figure 2d and Figure 3d. In both cQED quantum systems - the qubit is being driven by a square pulse; however the collapse operator is added on a different part of the system. In Figure 2d, the collapse operator is also added on the qubit. In Figure 3d, the collapse operator is added onto the cavity.

These differences account for the conclusion that the system with a collapse operator and drive on the qubit decays at a slower rate than the system which drives the qubit and collapses the cavity. The former is shown decaying to the ground state at a slower rate because Figure 2d contains faint pink rabi oscillations among a purple background. Figure 3d only contains purple and no rabi oscillation. Pink indicates an expectation value of 0.3-0.5. In this program, the expectation value is a projector of the excite state. A higher expectation value indicates a greater chance of the quantum system being in the excited state. Therefore, since Figure 2d contains higher expectation values, a quantum system that drives and collapses the qubit will remain in the excited state for a longer period.

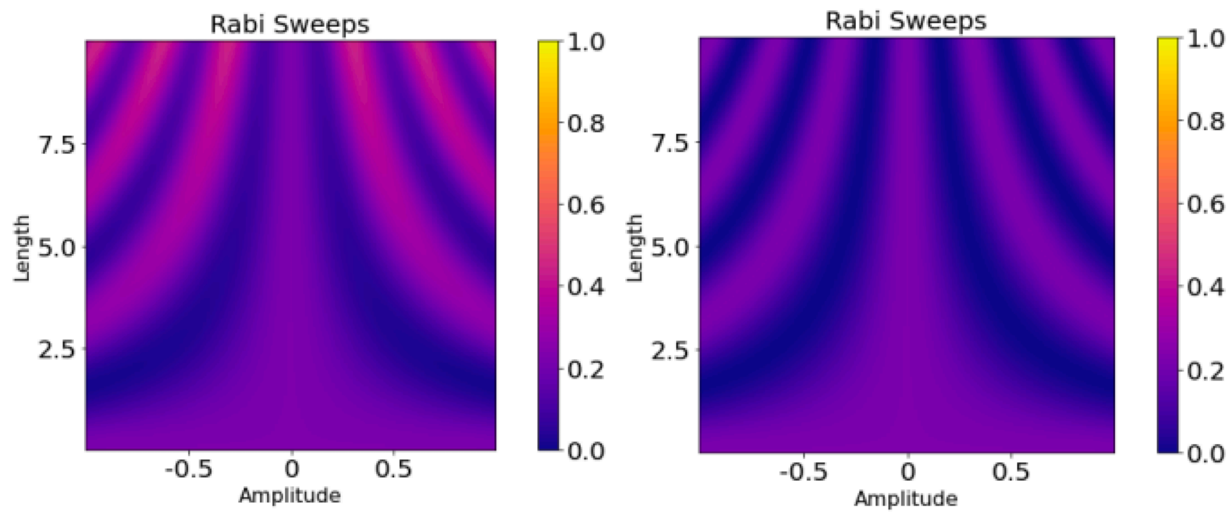

Figure 4: Rabi sweep of amplitude and length showing a quantum state that collapses and drives the qubit decays slower than a quantum state that collapses and drives two different parts of the system. The qubit, cavity, and interaction Jaynes-Cumming Hamiltonian are evolved by the Lindblad master equation. A different collapse operator is added in each case. Gamma and kappa are qubit and cavity dissipation rates.

(a) The qubit is driven by a square pulse and σ- collapse operator is added, Gamma = 0.1

(b) The qubit is driven by a square pulse and annihilation (a) collapse operator is added, Kappa = 0.1

In both Figure 4a and 4b, the cavity has a much higher frequency than the qubit. Both figures are measuring the expectation that the qubit is the excite state and the cavity is in the first Fock state. Additionally, the same amount of kappa/gamma (0.1) and square pulse is being applied to both cases. The main difference between the two figures is that the collapse operator is applied on a different part of the quantum system. In Figure 4a, a decay collapse operator is applied on the qubit while in Figure 4b a decay collapse operator is applied on the cavity.

Figure 4 also shares similar behavior to Figure 2 and Figure 3. Figure 4a which contains both the drive and collapse operators on the qubit decays at a slower rate than the system represented by Figure 4b. The system in Figure 4b drives the qubit while collapsing the cavity.

Conclusion

The Superconducting Qubits Training Program can be used to simulate an open quantum system and study decay behaviors with gamma and kappa values ranging from 0 to 1. In Figures 2 and 3, it was observed that a greater value of gamma or kappa led to a greater rate of decay from the excite to the ground state. This observation corroborated the expectation that gamma/kappa are dissipative rates of the qubit and cavity collapse operators. In Figure 4, it was also observed that a quantum system that collapses and drives the qubit will decay at a slower rate than a quantum system that drives the qubit but collapses the cavity. This observation highlights a distinction between the decay effect of driving and collapsing one part of the system in comparison driving and collapsing two different components.

Possible improvements to SQTP is the introduction of a lower temperature, projector to track the qubit and cavity state separately, and creating simulations that track the likelihood of the qubit being in the qubit excite state and the cavity in the second Fock state.

The Program

Here is a link to the code on GitHub: https://github.com/sblopez-coder/SweepsNoise

There is also a Sweeps Noise manual available that describes parts of the code in further detail.

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